Tuple

In [1]:
```
tup1 = ()
```

In [2]:
```
tup2 = (1,2,3)
```

In [3]:
```
tup3 = (10.77,30.6,40.55)
```

In [4]:
```
tup4 = ('one','two','three')
```

In [5]:
```
tup5 = ('somu',35,(24,56),(150,45))
```

In [6]:
```
tup6 = (100,'somu',17.654)
```

In [7]:
```
tup7 = ('asif',25,[50,100],[150,90],{'jhon','david'},(99,33,22))
```

In [8]:
```
len(tup7)
```

Out[8]:
```
6
```

Tuple indexing

In [9]:
```
tup2[0]
```

Out[9]:
```
1
```

In [10]:
```
tup4[0]
```

Out[10]:
```
'one'
```

In [11]:
```
tup4[0][0]
```

Out[11]:
```
'o'
```

In [12]:
```
tup4[-1]
```

Out[12]:
```
'three'
```

In [13]:
```
tup5[-1]
```

Out[13]:
```
(150, 45)
```

Tuple slicing

In [14]:
```
my_tuple = ('one' , 'two' , 'three' , 'four' , 'five' , 'six' , 'seven' , 'eight')
```

In [15]:
```
my_tuple[0:3]
```

Out[15]:
```
('one', 'two', 'three')
```

In [16]:
```
my_tuple[:]
```

Out[16]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

In [17]:
```
my_tuple.count('one')
```

Out[17]:
```
1
```

tuple object does not support item deletion and item assignment

Loop through a tuple

In [18]:
```
my_tuple
```

Out[18]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

In [19]:
```
for i in my_tuple:
    print(i)
```

```
one
two
three
four
five
six
seven
eight
```

In [20]:
```
for i in enumerate(my_tuple):
    print(i)
```

```
(0, 'one')
(1, 'two')
(2, 'three')
(3, 'four')
(4, 'five')
(5, 'six')
(6, 'seven')
(7, 'eight')
```

count function in tuple

In [21]:
```
tuple = (1,2,3,1,2,3,3,4,5)
```

In [23]:
```
print(tuple.count(1))
print(tuple.count(2))
print(tuple.count(3))
```

```
2
2
3
```

Index position in tuple

In [24]:
```
my_tuple
```

Out[24]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

In [26]:
```
print(my_tuple.index('one'))
print(my_tuple.index('two'))
print(my_tuple.index('five'))
```

```
0
1
4
```

Tuple membership

In [27]:
```
my_tuple
```

Out[27]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

In [29]:
```
'one' in my_tuple
```

Out[29]:
```
True
```

In [31]:
```
'ten' in  my_tuple
```

Out[31]:
```
False
```

In [34]:
```
if 'three' in my_tuple:
    print("three is present in mytuple")
else:
    print("three is not present in mytuple")
```

```
threes is present in mytuple
```

In [35]:
```
if 'eleven' in my_tuple:
    print("eleven is present in mytuple")
else:
    print("eleven is not present in mytuple")
```

```
eleven is not present in mytuple
```

Sorting in tuple

In [36]:
```
tuple20 = (20,30,40,32,56,1,2,3,24)
```

In [40]:
```
sorted(tuple20) #returms new sorted list does not change the orginal list
```

Out[40]:
```
[1, 2, 3, 20, 24, 30, 32, 40, 56]
```

In [39]:
```
sorted(tuple20,reverse= True) # it will returns the descending order
```

Out[39]:
```
[56, 40, 32, 30, 24, 20, 3, 2, 1]
```

Sets

In [42]:
```
''' 1.Unordered and unindexed collection of items.
    2.set elements are unique.duplicate elements are not allowed.
    3.set elements are immutable(cannot be changed)
    4.set itself is mutable.we can add or remove items from it'''
```

Out[42]:
```
' #1.Unordered and unindexed collection of items.\n    2.set elements are unique.duplicate el
ements are not allowed.\n    3.set elements are immutable(cannot be changed)\n    4.set itsel
f is mutable.we can add or remove items from it'
```

set creation

In [43]:
```
myset = {1,2,3,4,5}
```

In [44]:
```
myset
```

Out[44]:
```
{1, 2, 3, 4, 5}
```

In [45]:
```
len(myset)
```

Out[45]:
```
5
```

In [46]:
```
myset = {1,2,4,4,2,3}
```

In [47]:
```
myset
```

Out[47]:
```
{1, 2, 3, 4}
```

In [50]:
```
myset1 = {1.2,3.4,5.6,6.7}
myset2 = {'asif','prakesh','raju'}
myset3 = {10,20,'ramu',(11,22,33)}
```

In [52]:
```
myset4 = {1,2,3,[1,2,3]} #set does not allow mutable item
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[52], line 1
----> 1 myset4 = {1,2,3,[1,2,3]} #set does not allow mutable item

TypeError: unhashable type: 'list'
```

In [55]:
```
myset5 = set()
print(type(myset5))
```

```
<class 'set'>
```

In [56]:
```
myset1
```

Out[56]:
```
{1.2, 3.4, 5.6, 6.7}
```

Loop through set

In [57]:
```
myset1
```

Out[57]:
```
{1.2, 3.4, 5.6, 6.7}
```

In [58]:
```
for i in myset1:
    print(i)
```

```
1.2
3.4
5.6
6.7
```

In [59]:
```
for i in enumerate(myset1):
    print(i)
```

```
(0, 1.2)
(1, 3.4)
(2, 5.6)
(3, 6.7)
```

Set Membership

In [60]:
```
myset
```

Out[60]: {1, 2, 3, 4}

In [61]:
```
1 in myset
```

Out[61]: True

In [62]:
```
5 in myset
```

Out[62]: False

In [64]:
```
if 3 in myset: # Check if 'three' exist in the set
   print('3 is present in the set')
else:
   print('3 is not present in the set')
```

```
3 is present in the set
```

In [65]:
```
if 5 in myset: # Check if 'three' exist in the set
   print('5 is present in the set')
else:
   print('5 is not present in the set')
```

```
5 is not present in the set
```

Add and remove items

In [70]:
```
myset = {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

Add function

In [68]:
```
myset.add('Nine')
```

In [71]:
```
myset
```

Out[71]: {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}

update function

In [79]:
```
myset.update(['TEN','eleven','twelve'])
myset
```

Out[79]:
```
{'TEN',
 'e',
 'eight',
 'eleven',
 'five',
 'four',
 'l',
 'n',
 'one',
 'seven',
 'six',
 't',
 'ten',
 'three',
 'twelve',
 'two',
 'v',
 'w'}
```

Remove funtion

In [81]:
```
myset.remove('eight')
```

In [82]:
```
myset
```

Out[82]:
```
{'TEN',
 'e',
 'eleven',
 'five',
 'four',
 'l',
 'n',
 'one',
 'seven',
 'six',
 't',
 'ten',
 'three',
 'twelve',
 'two',
 'v',
 'w'}
```

In [84]:
```
myset
```

Out[84]:
```
{'TEN',
 'e',
 'eleven',
 'five',
 'four',
 'l',
 'n',
 'one',
 'seven',
 'six',
 't',
 'ten',
 'three',
 'twelve',
 'two',
 'v',
 'w'}
```

Clear

In [85]:
```
myset.clear()
myset
```

Out[85]:
```
set()
```

In [86]:
```
myset1
```

Out[86]:
```
{1.2, 3.4, 5.6, 6.7}
```

In [87]:
```
myset2 = {}
```

In [89]:
```
myset2 = myset1
myset2
```

Out[89]: {1.2, 3.4, 5.6, 6.7}

Set operations

In [90]:
```
A = {1,2,3,4,5}
B = {4,5,6,7,8}
C = {8,9,10}
```

union op

In [91]:
```
A | B
```

Out[91]: {1, 2, 3, 4, 5, 6, 7, 8}

In [92]:
```
A.union(B)
```

Out[92]: {1, 2, 3, 4, 5, 6, 7, 8}

update

In [93]:
```
A.update(B,C)
```

In [94]:
```
A
```

Out[94]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

Intersection

it will returns the common elements

In [95]:
```

A = {1,2,3,4,5}
B = {4,5,6,7,8}
```

In [96]:
```
A & B
```

Out[96]: {4, 5}

In [97]:
```
A.intersection_update(B)
```

In [98]:
```
A
```

Out[98]: {4, 5}

Difference

set of elements that are only in A but not in B

In [99]:
```

A = {1,2,3,4,5}
B = {4,5,6,7,8}
```

In [100…
```
A - B
```

Out[100… {1, 2, 3}

difference_update

In [103…
```
B.difference_update(A)
B
```

Out[103… {6, 7, 8}

other buitl in fuctions in set

In [104…
```
A = {1,2,3,4,5,6,7,8,9}
```

In [106…
```
print(sum(A))
print(min(A))
print(max(A))
print(len(A))
```

```
45
1
9
9
```

In [108…
```
list(enumerate(A))
```

Out[108… [(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8), (8, 9)]

In [109…
```
D = sorted(A,reverse = True)
D
```

Out[109… [9, 8, 7, 6, 5, 4, 3, 2, 1]

In [110…
```
sorted(D)
```

Out[110… [1, 2, 3, 4, 5, 6, 7, 8, 9]

Dictionary

In [111…
```
''' 1.Dictionary is a mutable data type in Python.
    2.A python dictionary is a collection of key and value pairs separated by a colon (:) & en
     in curly braces {}.
    3.Keys must be unique in a dictionary, duplicate values are allowed.'''
```

Out[111… ' 1.Dictionary is a mutable data type in Python. \n   2.A python dictionary is a collection of key and value pairs separated by a colon (:) & enclosed \n   in curly braces {}. \n 3.Keys must be unique in a dictionary, duplicate values are allowed.'

In [112…
```
mydict = dict() # empty dictionary
mydict
```

Out[112… {}

In [113…
```
mydict = {}
```

In [114…
```
mydict
```

Out[114… {}

In [115…
```
mydict = {1:'one' , 2:'two' , 3:'three'} # dictionary with integer keys
```

```
mydict
```

Out[115…
```
{1: 'one', 2: 'two', 3: 'three'}
```

In [116…
```
mydict = {1:'one' , 2:'two' , 3:'three'} # dictionary with integer keys
mydict
```

Out[116…
```
{1: 'one', 2: 'two', 3: 'three'}
```

In [117…
```
mydict = {1:'one' , 'A':'two' , 3:'three'} # dictionary with mixed keys
mydict
```

Out[117…
```
{1: 'one', 'A': 'two', 3: 'three'}
```

In [118…
```
mydict.keys()
```

Out[118…
```
dict_keys([1, 'A', 3])
```

In [119…
```
mydict.items()
```

Out[119…
```
dict_items([(1, 'one'), ('A', 'two'), (3, 'three')])
```

In [120…
```
mydict = {1:'one' , 2:'two' , 'A':['asif' , 'john' , 'Maria']} # dictionary with
mydict
```

Out[120…
```
{1: 'one', 2: 'two', 'A': ['asif', 'john', 'Maria']}
```

In [122…
```
mydict = {1:'one' , 2:'two' , 'A':['asif' , 'john' , 'Maria'], 'B':('Bat' , 'cat','hat')}
mydict
```

Out[122…
```
{1: 'one',
 2: 'two',
 'A': ['asif', 'john', 'Maria'],
 'B': ('Bat', 'cat', 'hat')}
```

In [123…
```
keys = {'a' , 'b' , 'c' , 'd'}

mydict3 = dict.fromkeys(keys) # Create a dictionary from a sequence of keys
mydict3
```

Out[123…
```
{'a': None, 'b': None, 'c': None, 'd': None}
```

In [124…
```
keys = {'a' , 'b' , 'c' , 'd'}
values = (10,20,30)
mydict3 = dict.fromkeys(keys,values) # Create a dictionary from a sequence of keys
mydict3
```

Out[124…
```
{'a': (10, 20, 30), 'b': (10, 20, 30), 'c': (10, 20, 30), 'd': (10, 20, 30)}
```

In [129…
```
values.append(40)
mydict3
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[129], line 1
----> 1 values.append(40)
      2 mydict3

AttributeError: 'tuple' object has no attribute 'append'
```

Asscessing items

In [130…

```
mydict = {1:'one' , 2:'two' , 3:'three' , 4:'four'}
mydict
```

Out[130…

```
{1: 'one', 2: 'two', 3: 'three', 4: 'four'}
```

In [132…

```
print(mydict[1])
print(mydict.get(1))
```

```
one
one
```

In [155…

```
mydict1 = {'Name':'Asif' , 'ID': 74123 , 'DOB': 1991 , 'job' :'Analyst'}
mydict1
```

Out[155…

```
{'Name': 'Asif', 'ID': 74123, 'DOB': 1991, 'job': 'Analyst'}
```

In [156…

```
mydict1['Name']
```

Out[156…

```
'Asif'
```

Add and remove changes

In [157…

```
mydict1
```

Out[157…

```
{'Name': 'Asif', 'ID': 74123, 'DOB': 1991, 'job': 'Analyst'}
```

In [158…

```
mydict1['DOB'] = 1992
mydict1['Address'] = 'Delhi'
mydict1
```

Out[158…

```
{'Name': 'Asif',
 'ID': 74123,
 'DOB': 1992,
 'job': 'Analyst',
 'Address': 'Delhi'}
```

In [159…

```
dict1 = {'DOB':1995}
mydict1.update(dict1)
mydict1
```

Out[159…

```
{'Name': 'Asif',
 'ID': 74123,
 'DOB': 1995,
 'job': 'Analyst',
 'Address': 'Delhi'}
```

In [160…

```
mydict1.popitem()
```

Out[160…

```
('Address', 'Delhi')
```

In [161…

```
del mydict1['DOB']
mydict1
```

Out[161…

```
{'Name': 'Asif', 'ID': 74123, 'job': 'Analyst'}
```

In [162…

```
mydict1.clear()
mydict1
```

Out[162…

```
{}
```

In [163…
```
del mydict1
mydict1
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[163], line 2
      1 del mydict1
----> 2 mydict1

NameError: name 'mydict1' is not defined
```

Copy dictionary

In [164…
```
mydict = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki'}
mydict
```

Out[164…
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

In [165…
```
mydict1 = mydict
```

In [166…
```
mydict1
```

Out[166…
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

In [167…
```
id(mydict1),id(mydict)
```

Out[167…
```
(2774709971136, 2774709971136)
```

In [169…
```
mydict2 = mydict.copy()
```

In [170…
```
mydict2
id(mydict2)
```

Out[170…
```
2774708116800
```

In [171…
```
mydict['address'] = 'mumbai'
```

In [172…
```
print(mydict1)
print(mydict)
```

```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki', 'address': 'mumbai'}
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki', 'address': 'mumbai'}
```

In [173…
```
mydict2
```

Out[173…
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

Loop through dictionary

In [175…
```
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki' }
mydict1
```

Out[175…
```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

In [176…
```
for i in mydict:
    print(i,':',mydict[i])
```

```
Name : Asif
ID : 12345
DOB : 1991
Address : Hilsinki
address : mumbai
```

In [177…

```python
for i in mydict:
    print(mydict[i])
```

```
Asif
12345
1991
Hilsinki
mumbai
```

Dictionsry membership

In [179…

```python
mydict1 = {'Name':'Asif' , 'ID': 12345 , 'DOB': 1991 , 'Address' : 'Hilsinki' }
mydict1
```

Out[179…

```
{'Name': 'Asif', 'ID': 12345, 'DOB': 1991, 'Address': 'Hilsinki'}
```

In [180…

```python
'Name' in mydict1
```

Out[180…

```
True
```

In [181…

```python
'somu' in mydict1
```

Out[181…

```
False
```

All/any

In [184…

```python
all(mydict1)
```

Out[184…

```
True
```

In [185…

```python
any(mydict1)
```

Out[185…

```
True
```

In [ ]: